Experiment 10
Zero-State and Zero-Input Response, Impulse Response and Convolution

/MATLAB Drive/Experiment_10_1.m

```
clc; clear; close all;
% Experiment 10: Zero-input, zero-state, impulse response and convolution
% System: dy(t)/dt + 2y(t) = x(t)
num = [1];
den = [1 2];
sys = tf(num, den);
disp('Transfer function:');
sys
disp('Pole:');
disp(pole(sys));
t = 0:0.01:5;
dt = t(2) - t(1);
h = exp(-2*t); % Impulse response for t >= 0
x_step = ones(size(t)); % Unit-step input
% Impulse response
figure;
impulse(sys, t);
grid on;
title('Impulse Response h(t)');
xlabel('Time (s)'); ylabel('h(t)');
% Zero-input response: y(0-) = 1, x(t) = 0
y_zi = exp(-2*t);
figure;
plot(t, y_zi, 'LineWidth', 1.5);
grid on;
title('Zero-Input Response');
xlabel('Time (s)'); ylabel('y_{zi}(t)');
% Zero-state response to unit step, with zero initial conditions
[y_zs, tout] = step(sys, t);
y_zs = squeeze(y_zs);
figure;
plot(tout, y_zs, 'LineWidth', 1.5);
grid on;
title('Zero-State Response to Unit Step');
xlabel('Time (s)'); ylabel('y_{zs}(t)');
```

```
% Numerical convolution approximation:
% y_zs(t) = integral from 0 to t of x(tau)h(t-tau) d(tau)
y_conv_full = conv(x_step, h) * dt;
y_conv = y_conv_full(1:length(t));
% Trapezoidal endpoint correction for the sampled convolution integral
y_conv = y_conv - 0.5*dt*(x_step(1)*h + x_step.*h(1));
% Total response for y(0-) = 1 and x(t) = u(t)
y_total = y_zi + y_zs(:).';
figure;
plot(t, y_zi, t, y_zs(:).', t, y_total, 'LineWidth', 1.3);
grid on;
legend('Zero-input', 'Zero-state', 'Total response', 'Location', 'best');
title('Comparison of Response Components');
xlabel('Time (s)'); ylabel('Response');
% Compare convolution result with MATLAB lsim zero-state response
y_lsim = lsim(sys, x_step, t);
figure;
plot(t, y_conv, '--', t, y_lsim(:).', '-');
grid on;
legend('Convolution approximation', 'lsim zero-state response', ...
    'Location', 'best');
title('Convolution Verification');
xlabel('Time (s)'); ylabel('y_{zs}(t)');

```

Code

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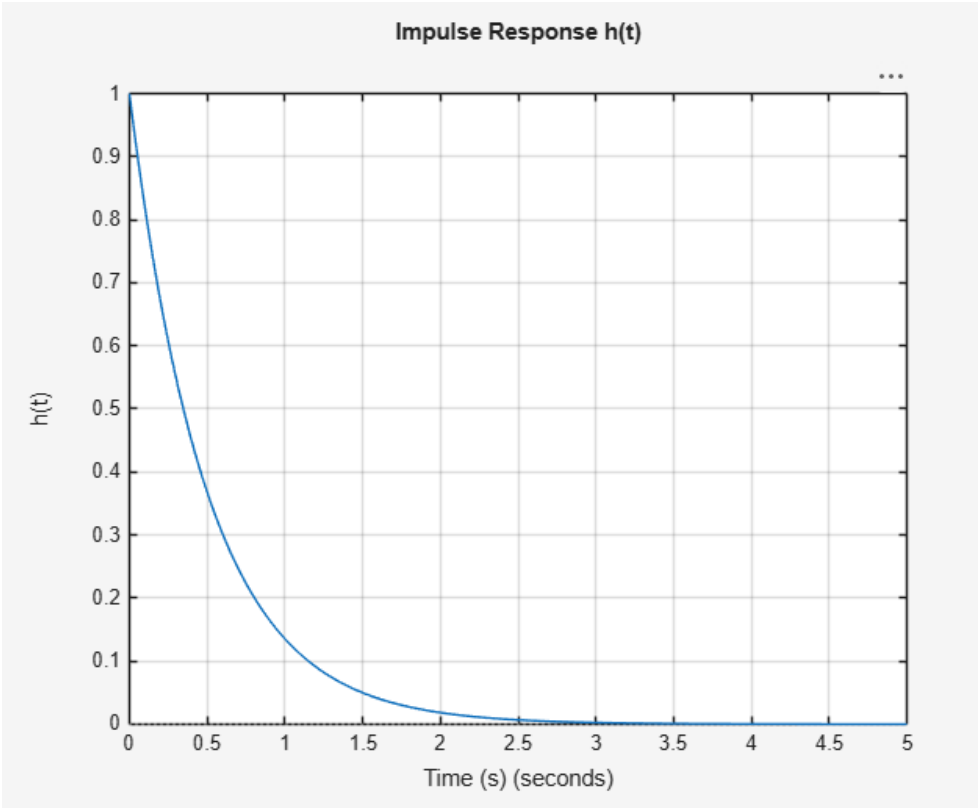


Figure 1

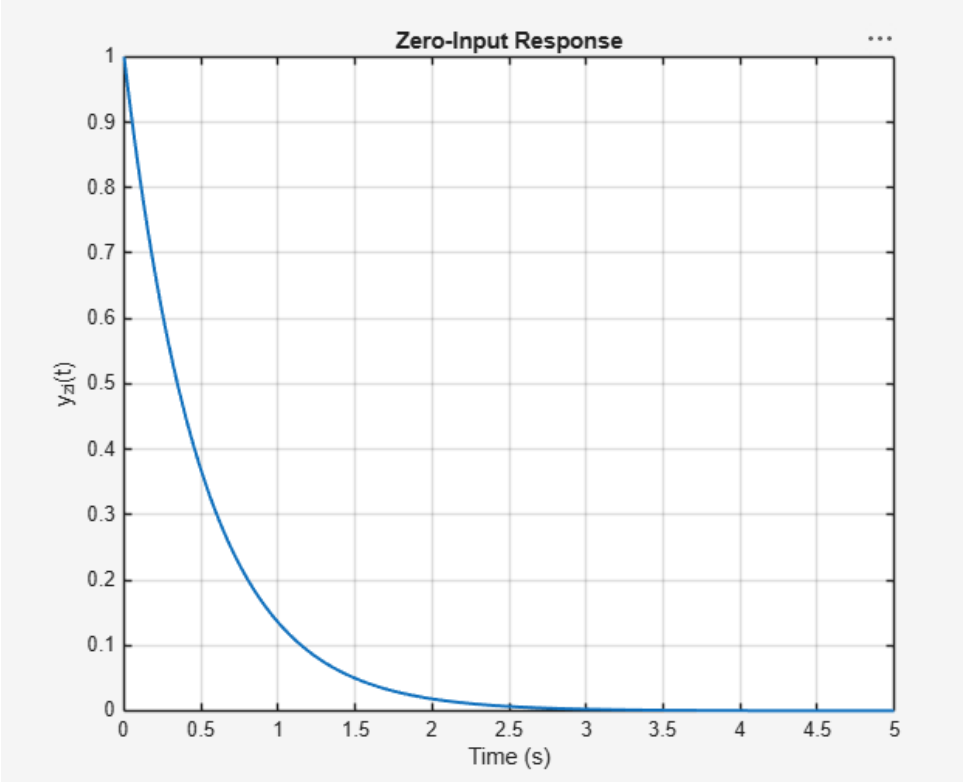


Figure 2

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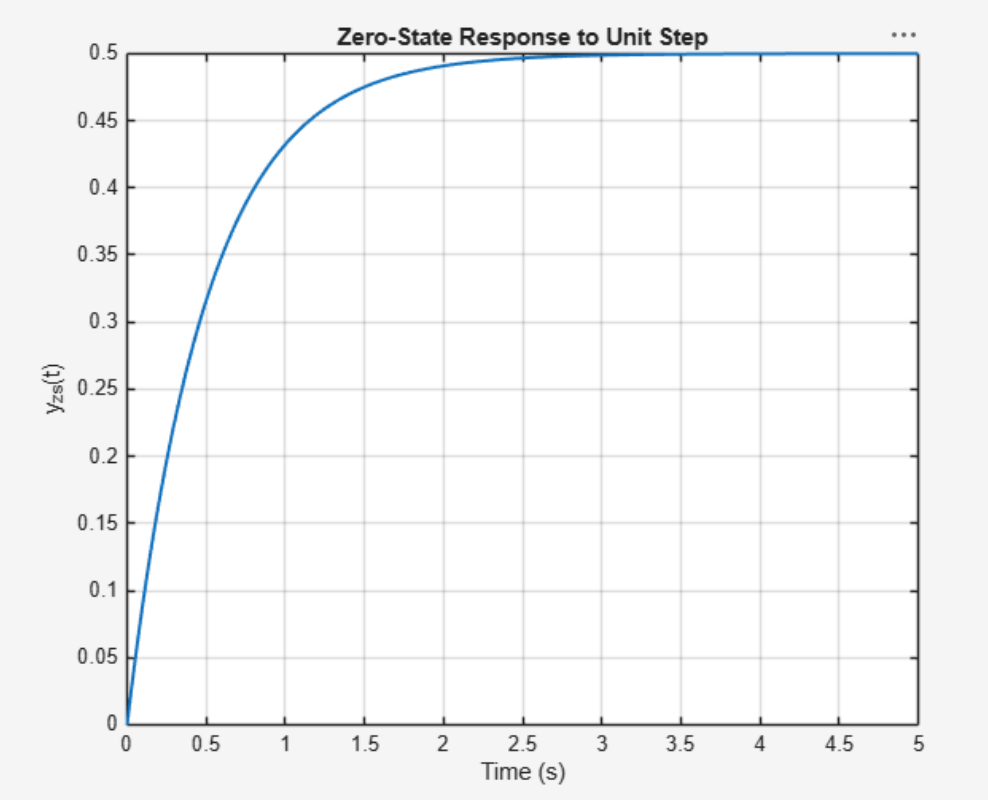


Figure 3

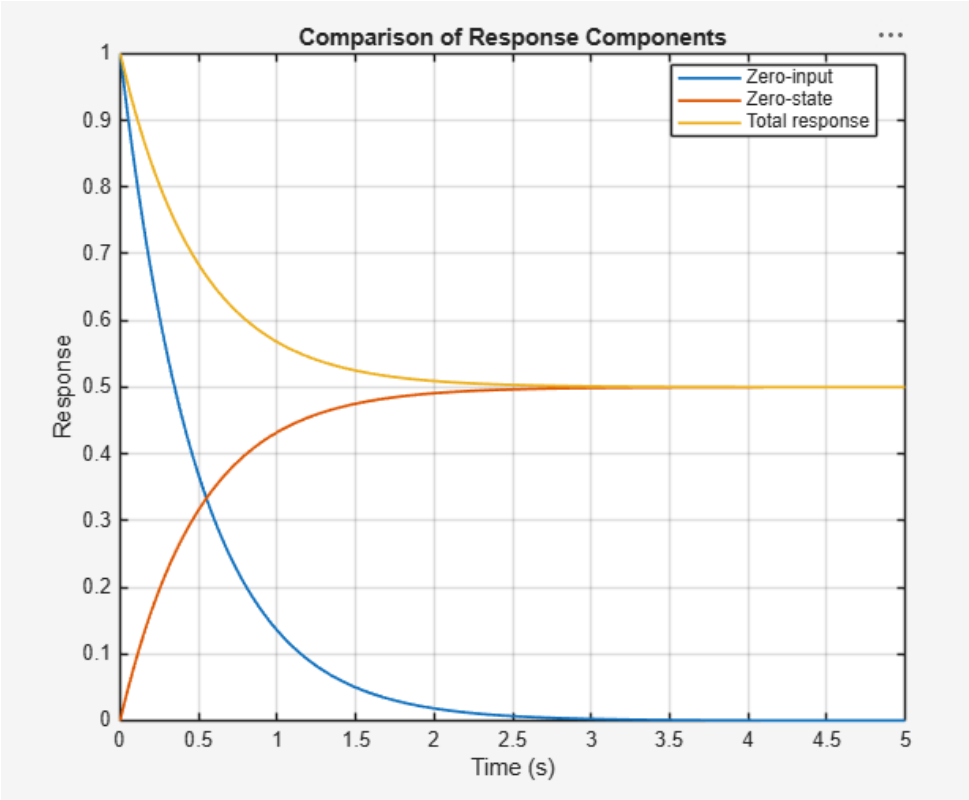


Figure 4

Experiment 10
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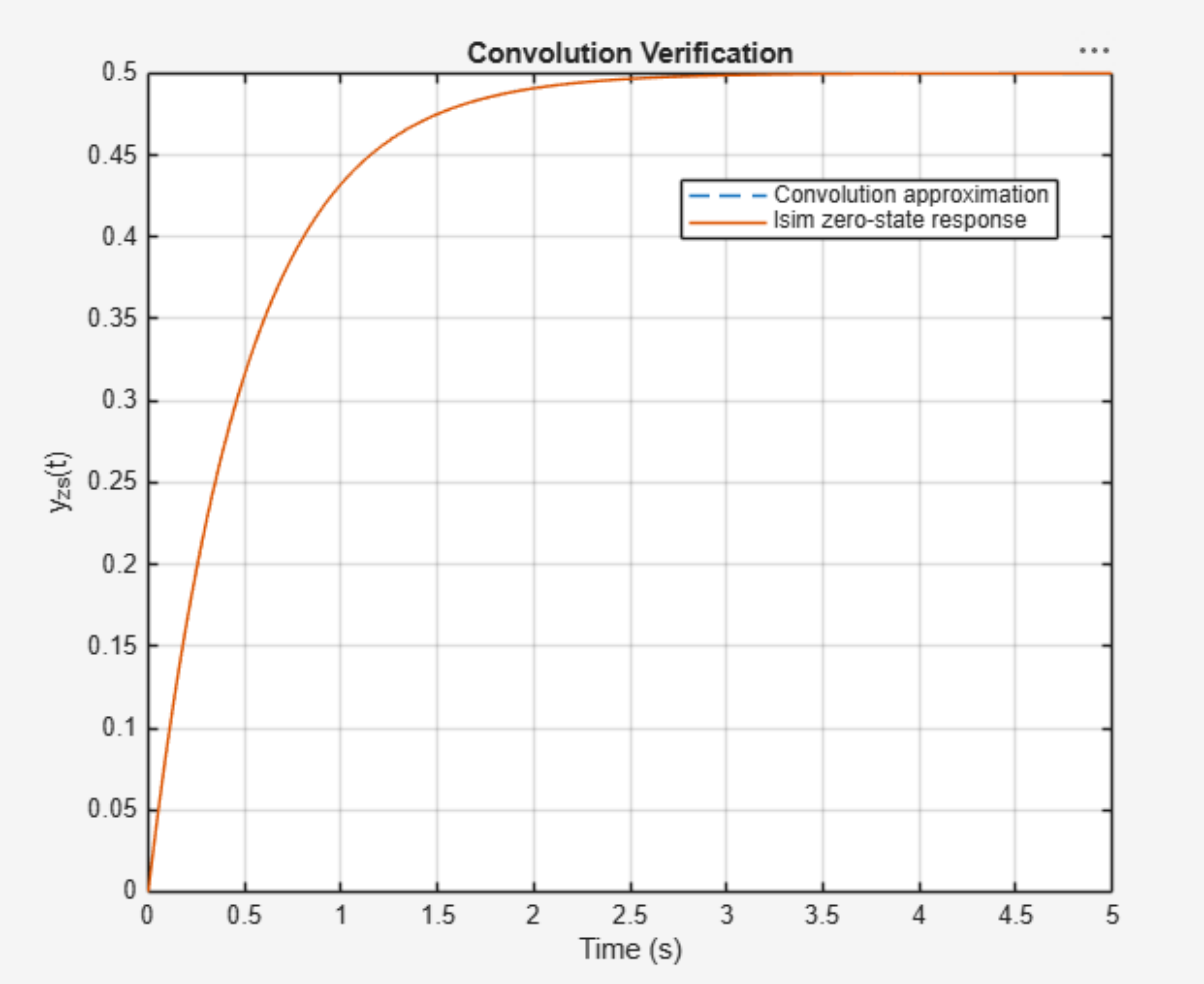


Figure 5